WASP-43b

R-0408

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_250212` · created 2026-07-19T05:21:22+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460718.829 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.057
Transit depth (Rp/Rs)^2	3.92 +/- 2.24 [%]
Orbital Inclination (inc)	87.11 +/- 2.84
Airmass coefficient 1 (a1)	0.151 +/- 0.037
Airmass coefficient 2 (a2)	1.09 +/- 0.21
Scatter in the residuals of the lightcurve fit is	19.55 %
Best Comparison Star	None
Optimal Aperture	2.93
Optimal Annulus	10.69
Transit Duration (day)	0.0613 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.198 ± 0.057 vs 0.1594 (deviation 0.53σ)
Duration fit vs published	0.0613 d vs 0.0512 d (deviation 1.63σ)
Mid-time O–C	-6.1 min
Depth SNR	2.7
Residual scatter	19.55 %

Light curve

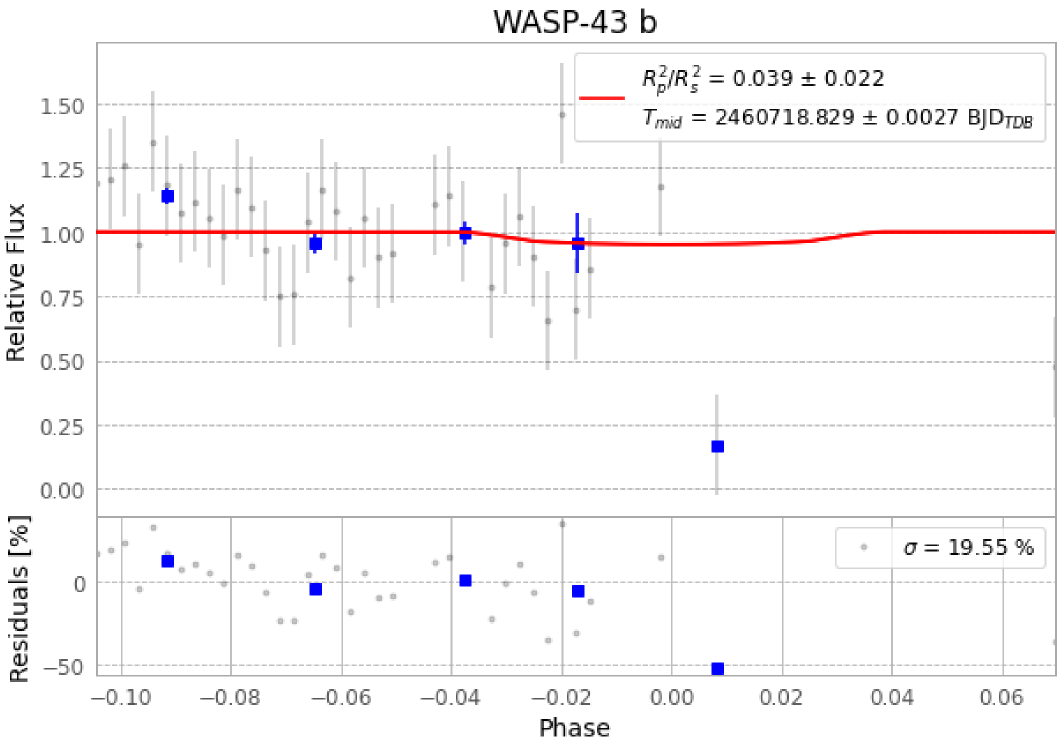

Figure 1 — FinalLightCurve_WASP-43 b_2025-02-12.png (SHA-256 7ed08ec4bcf4defe…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [283, 403], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 09a488f3ee14c042…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80`

Data provenance

76 FITS frames (50 MB), `WASP-43250212054217.FITS` → `WASP-43250212092716.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-250212.log (0133a163a085…), FinalLightCurve_WASP-43 b_2025-02-12.png (7ed08ec4bcf4…), FinalLightCurve_WASP-43 b_2025-02-12.csv (1a3a165e30c9…)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)